Ian Kronquist

iankronquist@gmail.com
+1 971 222 5148 • github.com/iankronquist
C, C++, Python, Rust, Windows, Linux, Hypervisors, Kernels

Employment

Microsoft

Platform Security & Kernel Security Teams *Software Engineer II, September 2017-Present*
Designed and implemented mitigations to harden Windows components and break exploit techniques. One such mitigation, Win32k Object Type Isolation, successfully defended against over a dozen CVE-worthy cases in 2019 alone. Developed CastGuard, a compiler based mitigation to prevent illegal downcasts in C++. Helped deliver Address Sanitizer support for the Microsoft MSVC C++ compiler. Became the maintainer of the Windows heap used throughout the OS and improved heap performance and debuggability.

Barkly Protects, Protection Team *Software Engineer, July 2016-July 2017*
Core developer of a Windows filesystem filter driver and a hypervisor designed to detect and stop malware. Collaborated with malware researchers to develop new rules to stop ransomware and measure protection coverage using the Cuckoo malware analysis framework.

Oregon State University Open Source Lab *Developer, November 2013-June 2016*
Wrote a web back end for the lab's NodeJS time management API. Created Docker development environments and Chef DevOps cookbooks for deploying the lab's Python web applications.

Puppet Labs, Puppet Language Design Team *Intern, June-September 2015*
Implemented a tool to do type checking and generate HTML documentation from code comments. Designed and prototyped a system for instrumenting the Puppet parser to extract test coverage statistics. Implemented built-in functions for the Puppet catalog compiler.

Mozilla, SUMO Engineering Team *Intern, June-September 2014*
Designed and implemented a Bayesian spam and abuse detection and filtering library in Python for Mozilla's user input Django site.

Reach Technology *Intern, July-September 2013*
Wrote a cross-platform C++ desktop application for the customer development kit for an embedded graphics display module.

Education

Oregon State University *Corvallis, Oregon September 2013-June 2017*
Bachelor of Science, Computer Science, Honors Degree. Implemented a honeypot to monitor vulnerable containers as a Senior Project.

AFS Turkey Foreign Exchange Program *Adana, Turkey September 2012-July 2013*
Attended Gündoğdu Koleji (Sunrise High School) in Adana, Turkey. Proficient in the Turkish Language.

Eagle Scout
Designed, constructed, and gathered donations for a lending library as an Eagle Project.

Talks

The Life & Death of Kernel Object Abuse by Type Isolation *OPCDE Dubai & Hack In the Box Amsterdam*